

DANIEL MATHEWS

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EDUCATION

University of California - San Diego

Junior Standing - Graduating in June 2025

Bachelor's, Data Science

- B.S. in Data Science (Specialization in Machine Learning / Artificial Intelligence)

PROFESSIONAL EXPERIENCE

Boeing

Huntington Beach, CA

Data Science Summer Intern

June 2024 – August 2024

- Collaborated with Boeing mentors to develop a Python-based visualization tool using NumPy, pandas, Matplotlib, Flask, and JSON that converted 5D hyperspherical flight trajectory data into accurate 2D dashboards for engineers using nested heatmaps.
- Led team as Integrated Product Leader in programming ML-driven autonomous Donkey Cars for a company-wide race challenge.

Data Science Society Software Team

La Jolla, CA

Board Member

October 2025 – Present

- Promoted to Project Manager after one month as a developer; organized dev team and stakeholder meetings to fix integration and ship a new DS3 consulting site.
- Used Github and Notion for team coordination, working with JavaScript (Node, React), Tailwind CSS, and HTML to execute and enhance full-stack projects.

Sprouts Farmers Market

Yorba Linda, CA

Courtesy Clerk

December 2023 – September 2024

- 100+ shoppers interacted with every shift; delivered attentive customer service by answering questions, suggesting products, and resolving concerns, youngest hire in branch history. Bagged groceries & organized carts, returns, spoiled goods, & cleaned messes/spills to support safe store operations.

PROJECTS & OUTSIDE EXPERIENCE

SKIPREDICT – Skipability Score of Any Given Song

Yorba Linda, CA

Predictive Algorithms, Responsive Design, Full-Stack Application

July 2025 – September 2025

- Engineered a **full-stack predictive model** to estimate **probability** that a user skips a song by developing a custom machine learning algorithm that analyzes and weights key audio features (energy, valence, tempo, ect).
- Tech Stack including **Python/Flask**, integrating the iTunes Search **API** with **JavaScript, HTML, and CSS** for an interactive frontend, **RESTful** endpoints + JSON data parsing,
- [Link to project](#)

Quantum Dimensionality Reduction for Portfolio Optimization

La Jolla, CA

PCA, Ridge Regression, Gradient Boosting, Feature Extraction

September 2025 - Present

- An algorithmic trading system that combines **dimensionality reduction techniques** (including a quantum-inspired approach) with **machine learning classifiers** to forecast price movements of the **NVIDIA stock** using **scikit-learn, Matplotlib, Seaborn, numpy, pandas**, and the **Yahoo Finance Library (yfinance)**
- [Link to project](#)

NFL Red Zone Play Sheet – Optimal Red Zone Plays Dashboard

La Jolla, CA

Bayesian Statistics + Simulations, Kinematic Analysis, Interactive Visualization

November 2025 – December 2025

- Interactive dashboard using **Python** (Pandas, NumPy, Matplotlib, SciPy) and **Bayesian Beta / Monte Carlo** simulations to evaluate route / formation patterns in red zone.
- Incorporates **defender positional analysis** and **reliability scores** to create a recommendation engine returning the highest expected success rate play formations + confidence.
- [Link to project](#)

CERTIFICATIONS

- Google Data Analytics Professional Certificate (Coursera) // Excel Desktop (Microsoft 365) Certificate // Data Structures in Python (LinkedIn Learning)
- Python and Statistics for Financial Analysis (Hong Kong University of Science and Technology) // SQL for Data Analysis (LinkedIn Learning)
- Create an Agent Using Agentforce DX Badge (Salesforce) // Python for Data Science and Machine Learning (LinkedIn Learning)

TECHNICAL SKILLS

- Frontend Development:** HTML, JavaScript, CSS, TypeScript, Eclipse IDE, React.js
- Backend & Databases:** REST APIs, SQL (MySQL, PostgreSQL), Node.js
- Programming & Data:** Python (pandas, NumPy, scikit learn, ect), R, ETL/ELT workflows, Java, C++
- Visualization:** Plotly, Tableau, Power BI